

RICHARD SO

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EDUCATION

Georgia Institute of Technology

08/2024 - 05/2025

M.S. Computer Science, Machine Learning - GPA: 4.0

Atlanta, GA

- Coursework: ML, Deep Learning, Computer Vision, NLP, Databases, Networks, Algorithms Honors

Georgia Institute of Technology

08/2021 - 05/2024

B.S. Computer Science - GPA: 4.0

Atlanta, GA

WORK EXPERIENCE

Amazon | Software Development Engineer

08/2025 - Present

- Maintained **data lake infrastructure** (AWS Glue, EMR, Lambda, S3) for Amazon Brand Analytics, enabling sellers to monitor **sales performance insights across 2B+ products**.
- Orchestrated **high-throughput data vending pipelines** with Apache Airflow and Spark to transform **~100 TB/week** of raw purchase activity into curated analytical datasets for internal teams and sellers.
- Developed core components of an **explainable ML root cause analysis service**, leveraging **Shapley value feature attribution** to identify drivers of product underperformance relative to market percentiles.

Data To Insights (D2I) Lab @ Georgia Tech | Research Assistant

01/2025 - 05/2025

- Investigated latency reduction in **multi-tenant LLMs for RAG workloads** by overlapping document retrieval and prefill.
- Led **scenario-based benchmarking** by simulating web crawler and ANNS retrieval to evaluate latency improvements of our custom token prefilling mechanism under realistic workloads.
- Analyzed results for and **co-authored MLSys research paper** demonstrating up to **11x speedup** with our strategy.

Amazon Web Services | Software Engineering Intern (ML)

05/2024 - 08/2024

- Implemented **ML-driven analysis of developer workflows** by aggregating behavioral features (keystrokes, UI events, time-on-task), enabling quantitative DX comparison across cloud platforms and **influencing AWS roadmap decisions**.
- Re-architected a batch inference pipeline to exploit parallelism, **reducing runtime by 85% (minutes → seconds)**.
- Automated developer intent and task classification from desktop screenshots using AWS **Rekognition**, **Textract**, and **multi-modal LLM prompting on Bedrock**.

Tanium | Software Engineering Intern

06/2023 - 08/2023

- Built CRUD logging to Tanium console's PostgreSQL backend and REST endpoints to support a **customer-facing audit feature**, allowing review of console activity and detection of unauthorized configuration changes.
- Resolved **50+ feature/bug tickets** within a 10-week internship maintaining a Knex.js + React TypeScript codebase.
- Ensured code quality and correctness by applying **TDD & validation best practices** using Jest, Jasmine, and Joi.

PROJECTS

Generative Data Augmentation for Image Classification 🌀

04/2024

- Experimented with **Stable Diffusion** and **ControlNet** to enhance image classification accuracy when data is scarce.
- Observed a **10% F1 increase** for Resnet-50 on a limited dataset when augmented with ControlNet-generated images.
- Orchestrated **large-scale DL experiments**, sweeping across 100+ hyperparameter and model configurations.

LC3Tools 🌀

09/2023 - 05/2024

- **Lead maintainer** of the educational tooling suite for coding, assembling, and simulating **LC-3** assembly programs.
- Actively collected student & instructor feedback to continuously drive **major quality-of-life enhancements**.
- Served **1000+ students every semester** as one of the core, required tools for Georgia Tech's CS2110 course.

SKILLS

Programming Languages | Python, TypeScript/JavaScript, Java, Scala, C/C++, Go, Lua

Frameworks & Libraries | NumPy, Pandas, SkLearn, PyTorch, Lightning, Apache Spark, React, Svelte

Databases & Misc. | PostgreSQL, SQLite, DynamoDB, Elasticsearch, Docker, Apache Airflow, AWS CDK